Markovian Arrival Process (MAP) Structure Types

A Reference Guide to the 19 Supported MAP Families

A Markovian Arrival Process (MAP) is a versatile stochastic model for arrival streams that generalises the Poisson process. It is characterised by a pair of matrices (D0, D1), both of size n x n, where n is the number of phases. D0 governs transitions that occur without an arrival, while D1 governs transitions that coincide with an arrival. The sum Q = D0 + D1 is the generator of the background continuous-time Markov chain (CTMC).

Key properties used throughout this document:
- rho1: first-lag autocorrelation of inter-arrival times.
- SCV: squared coefficient of variation of the marginal inter-arrival distribution.
- Renewal process: rho1 = 0 (consecutive inter-arrivals are independent).

Each section includes a sparsity visualisation: a grid where filled cells indicate non-zero entries and empty cells indicate zeros. Dark navy marks the D0 diagonal (always non-zero, always negative), blue marks D0 off-diagonal non-zeros, and green marks D1 non-zeros. 19 MAP families are described.

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1. mmpp

Zone / Category: Standard

Autocorrelation (rho1): Moderate (depends on structure)SCV range: >= 1

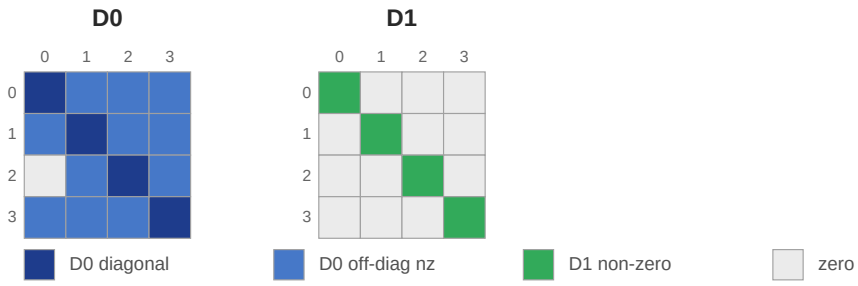
Description

Markov-Modulated Poisson Process. The background state evolves as a continuous-time Markov chain (CTMC), and arrivals occur at a Poisson rate that depends on the current state. D1 is diagonal, meaning arrivals do not trigger a state transition.

Matrix Structure

D0 = Q - diag(lambda): sparse/Coxian/hyper-Erlang generator minus arrival rates on the diagonal. Off-diagonal entries of D0 are a subset of Q. D1 = diag(lambda): diagonal matrix of per-state arrival rates; all off-diagonal entries are zero.

Sparsity Pattern (small example)



2. alternating_blocks

Zone / Category: Standard

Autocorrelation (rho1): Typically negative (block alternation) SCV range: Variable

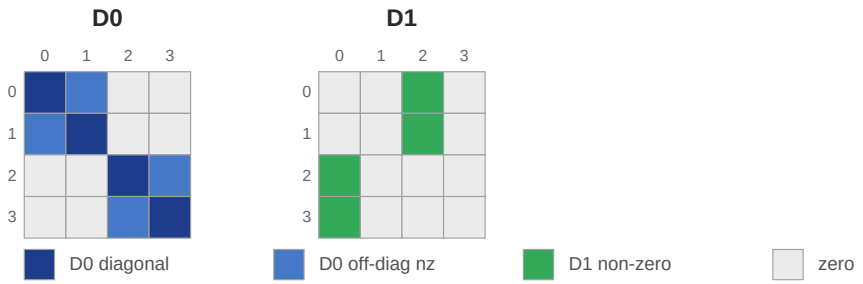
Description

Two-block MAP where each observed arrival switches the process to the first phase of the opposite block. Block A is the slow block and block B the fast block. Each block independently uses a sparse, Coxian, or hyper-Erlang internal structure.

Matrix Structure

D0 is block-diagonal: upper-left (block A) and lower-right (block B) are dense, all cross-block entries are zero. D1 is block off-diagonal: upper-right (A arrivals enter B) and lower-left (B arrivals enter A) are non-zero; the diagonal blocks of D1 are zero.

Sparsity Pattern (small example)



3. alternating_extreme

Zone / Category: Standard

Autocorrelation (rho1): Strongly negative (approaching -1) **SCV range: Approx. 1 (low to moderate)**

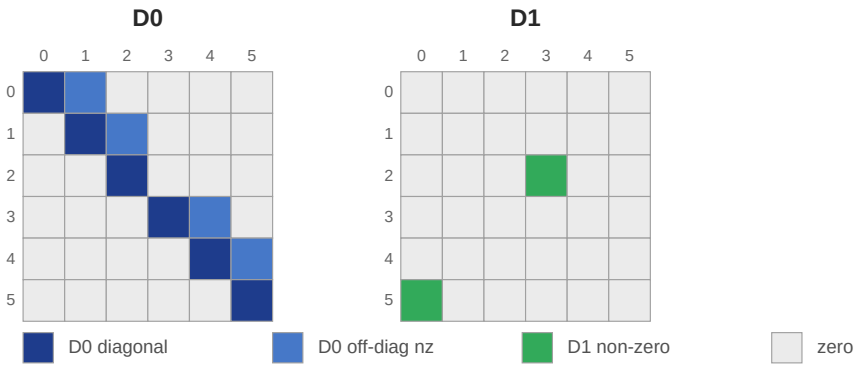
Description

Extreme slow/fast alternating MAP. Each block is an Erlang chain: the process advances through phases without arrivals, and only the final phase emits an arrival that starts the opposite block. The slow block has mean 8-80x longer than the fast block, creating very strong negative first-lag autocorrelation.

Matrix Structure

D0 is block-tridiagonal: within each Erlang chain, only the superdiagonal is non-zero (forward phase progression). All cross-block entries of D0 are zero. D1 has exactly two non-zero entries: D1[last_A, first_B] and D1[last_B, first_A]. All other D1 entries are zero.

Sparsity Pattern (small example)



4. alternating_extreme_hyper

Zone / Category: Standard

Autocorrelation (rho1): Negative (preserved from alternation) SCV range: High (hyper-Erlang branches widen SCV)

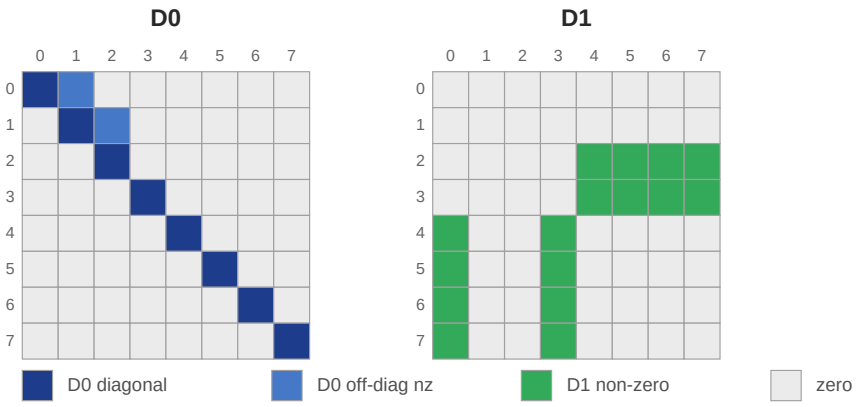
Description

Strict slow/fast alternation with hyper-Erlang branch variability. Like alternating_extreme but each block consists of multiple Erlang branches with different means. This widens the SCV, skewness, and kurtosis while preserving the negative autocorrelation from the alternation structure.

Matrix Structure

D0 is block-diagonal with each block having an internal hyper-Erlang structure (Erlang chains in parallel). D1 is block off-diagonal: only the last phase of each branch in block A can trigger an arrival into block B, and vice versa. Most entries of D1 are zero.

Sparsity Pattern (small example)



5. diverse_ph

Zone / Category: Renewal (rho1 = 0)

Autocorrelation (rho1): 0 (renewal process) **SCV range: Variable**

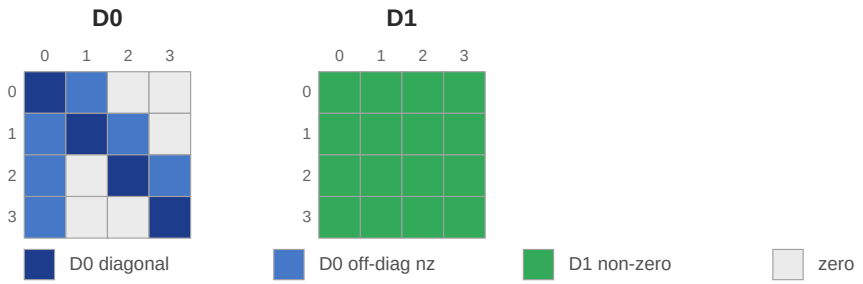
Description

Renewal MAP built from a phase-type (PH) distribution. Since each inter-arrival time is drawn independently from the same PH distribution, consecutive inter-arrivals are uncorrelated (rho1 = 0 exactly). Covers four PH sub-structures: Coxian, Hyper-Erlang, General (sparse), and Modified Hyper-Erlang.

Matrix Structure

D0 = T (PH subgenerator): sparsity depends on the PH type (Coxian = upper-triangular with resets; hyper-Erlang = block structure). D1 has rank 1: D1 = outer(t, alpha), so every row of D1 that corresponds to a state with a non-zero exit rate t_i is entirely non-zero, and rows with t_i = 0 are entirely zero.

Sparsity Pattern (small example)



6. modified_renewal_ph

Zone / Category: Standard

Autocorrelation (rho1): Positive or negative (targeted restart) SCV range: Variable (PH-dependent)

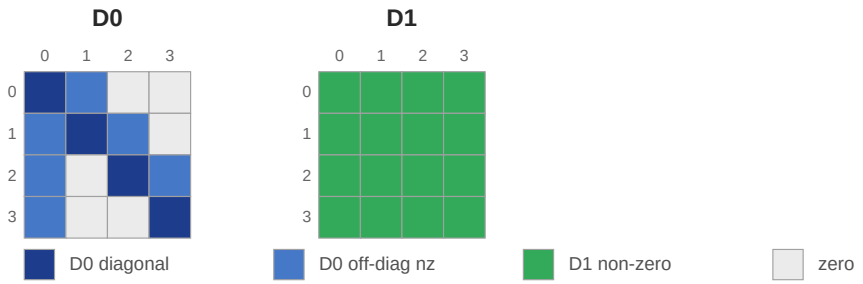
Description

PH-based MAP with a targeted (non-renewal) arrival restart distribution in D1. Unlike diverse_ph, the restart probabilities after an arrival are biased to induce positive or negative inter-arrival dependence. The sign of the correlation is controlled explicitly.

Matrix Structure

D0 = T (PH subgenerator): same sparse structure as diverse_ph. D1 is no longer rank-1: each row of D1 has its own non-zero pattern, biased so that exits from fast states preferentially restart in fast states (positive correlation) or slow states (negative correlation). Most rows remain dense.

Sparsity Pattern (small example)



7. random_map

Zone / Category: Standard

Autocorrelation (rho1): Random (no control) **SCV range: Random**

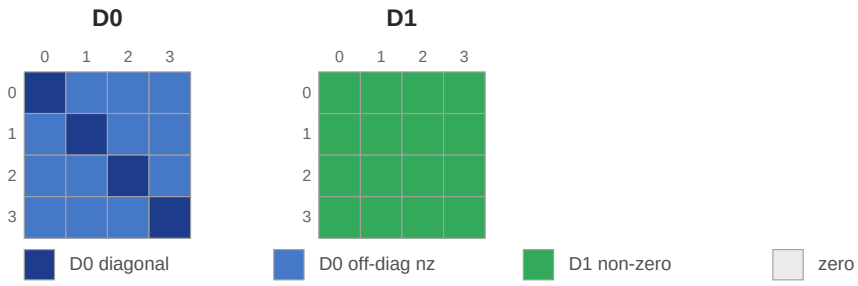
Description

Fully random MAP generated by the random_map() helper. Both D0 and D1 are constructed from random non-negative matrices subject to MAP validity constraints (D0 + D1 is a valid generator). Provides unstructured baseline coverage of the (rho1, SCV) space.

Matrix Structure

D0 has non-positive diagonal and non-negative off-diagonal entries; D1 has non-negative entries. Both matrices are generally dense: almost all entries are non-zero. D0 + D1 sums to a valid CTMC generator (rows sum to 0).

Sparsity Pattern (small example)



8. random_map_negative1

Zone / Category: Standard

Autocorrelation (rho1): Biased negative **SCV range: Random**

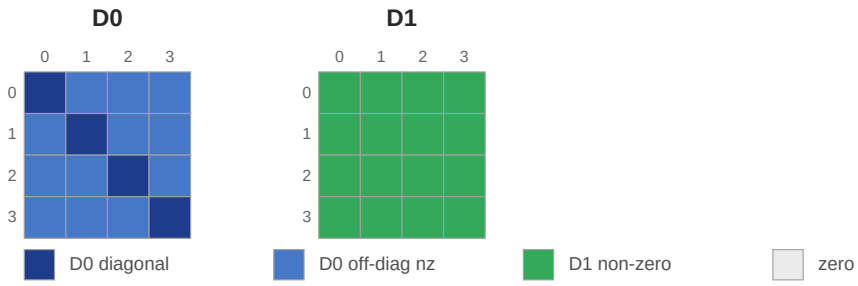
Description

Random MAP generated by the random_map_negative1() helper, which is biased to produce negative first-lag autocorrelation. Provides unstructured baseline coverage of the negative-rho1 region.

Matrix Structure

Similar matrix structure to random_map: both D0 and D1 are generally dense. The construction biases D1 toward off-diagonal dominant entries, encouraging state alternation after each arrival and thus negative autocorrelation, but the exact sparsity pattern is random.

Sparsity Pattern (small example)



9. erlang_renewal

Zone / Category: Renewal (rho1 = 0)

Autocorrelation (rho1): 0 (renewal process) **SCV range: 1/k (low for large k)**

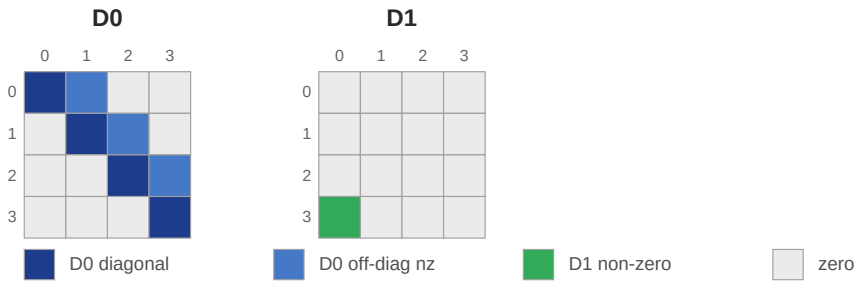
Description

Erlang-k renewal MAP. Each inter-arrival time is the sum of k i.i.d. exponential phases, giving SCV = 1/k. For k=1 this reduces to a Poisson process (SCV = 1). As k increases, inter-arrivals become increasingly deterministic. All consecutive inter-arrivals are independent (rho1 = 0).

Matrix Structure

D0 is tridiagonal: diagonal entries are -rate and superdiagonal entries are +rate (phase progression). All other D0 entries are zero. D1 has exactly one non-zero entry: D1[k-1, 0] = rate (last phase triggers arrival and restarts at phase 0). All other D1 entries are zero.

Sparsity Pattern (small example)



10. sparse_ph_renewal

Zone / Category: Renewal (rho1 = 0)

Autocorrelation (rho1): 0 (renewal process) **SCV range: Variable (general PH)**

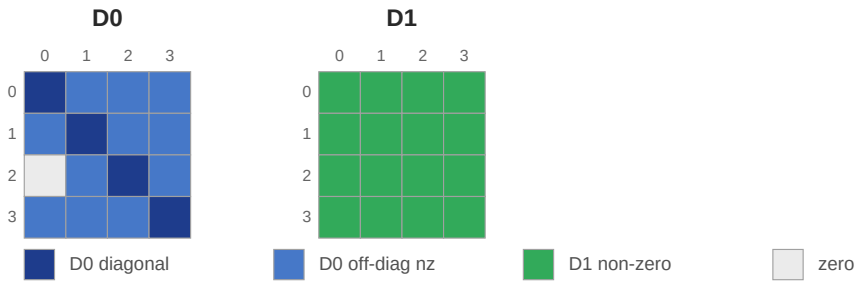
Description

Renewal MAP from a sparse general PH distribution. The PH subgenerator has many zero off-diagonal entries (only a guaranteed ring structure plus random non-zero edges). This gives a varied SCV range depending on the sampled PH. Inter-arrivals are always independent (rho1 = 0 exactly).

Matrix Structure

D0 = T (sparse PH subgenerator): has a guaranteed ring of forward transitions plus a random subset of additional off-diagonal entries; most off-diagonal entries are zero. D1 = outer(t, alpha): same rank-1 structure as diverse_ph. Non-zero rows of D1 correspond to states with non-zero exit rates.

Sparsity Pattern (small example)



11. hyper_erlang_renewal

Zone / Category: Renewal (rho1 = 0)

Autocorrelation (rho1): 0 (renewal process) **SCV range: High (multiple Erlang branches)**

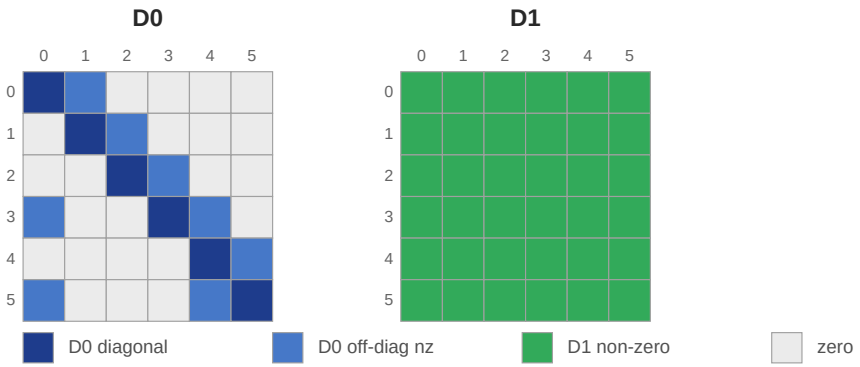
Description

Renewal MAP from a hyper-Erlang PH distribution. Multiple parallel Erlang chains with different rates are entered with probabilities drawn from a Dirichlet. This can achieve high SCV (>> 1) while remaining a renewal process (rho1 = 0 exactly).

Matrix Structure

D0 is block-structured: within each Erlang branch, only the superdiagonal is non-zero (phase progression); there are no cross-branch entries in D0. D1 is sparse: only the last phase of each branch has non-zero entries (the exit phases), and each such row is fully non-zero (rank-1 structure).

Sparsity Pattern (small example)



12. modified_hyper_erlang_renewal

Zone / Category: Renewal (rho1 = 0)

Autocorrelation (rho1): 0 (renewal process) **SCV range: High (wider than standard hyper-Erlang)**

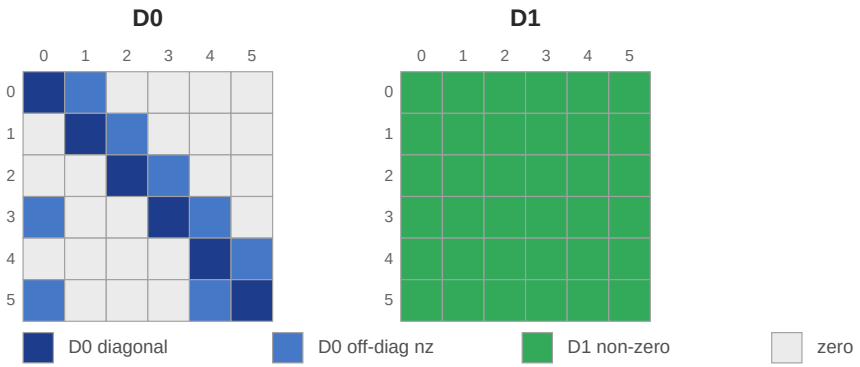
Description

Renewal MAP from a modified hyper-Erlang PH, where the restart distribution within each branch end is random (not simply returning to the branch start). This gives wider coverage of skewness and kurtosis than standard hyper-Erlang while keeping rho1 = 0 exactly.

Matrix Structure

D0 has the same hyper-Erlang block structure as hyper_erlang_renewal: within-branch superdiagonal only, no cross-branch entries. D1 has the same rank-1 outer-product structure, so non-zero rows appear at branch exits. The difference lies in the random entry distribution within those rows, not in the sparsity pattern.

Sparsity Pattern (small example)



13. slow_switching_mmpp

Zone / Category: Zone A: strong positive rho1

Autocorrelation (rho1): 0.5 to 0.95 (very positive) **SCV range: Moderate to high**

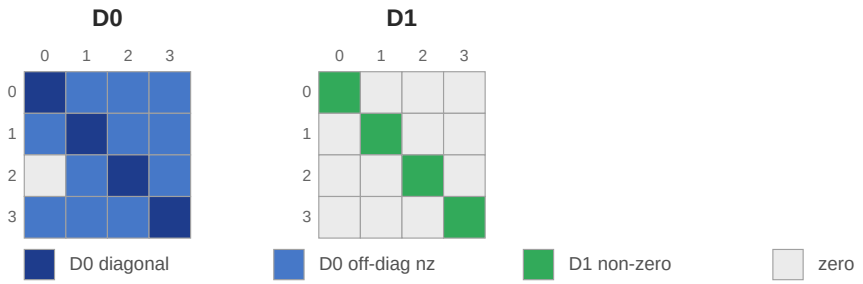
Description

MMPP with a slow background Markov chain. The switching rate between states is scaled down by a factor (0.001-0.15) relative to the arrival rates. Because the state persists across many consecutive arrivals, consecutive inter-arrival times are highly correlated. Rate spread across states independently controls SCV.

Matrix Structure

D1 = diag(lambda): diagonal only, identical structure to a standard MMPP. D0 = Q_slow - diag(lambda): the off-diagonal entries of D0 come from Q_slow (sparse CTMC scaled by switching_scale), all very small in magnitude relative to the diagonal. Sparsity pattern is the same as mmpp; only the magnitude ratio differs.

Sparsity Pattern (small example)



14. erlang_modulated

Zone / Category: Zone B: positive rho1, low SCV

Autocorrelation (rho1): > 0 (0.3-0.7 typical) **SCV range: < 1 (low, Erlang-controlled)**

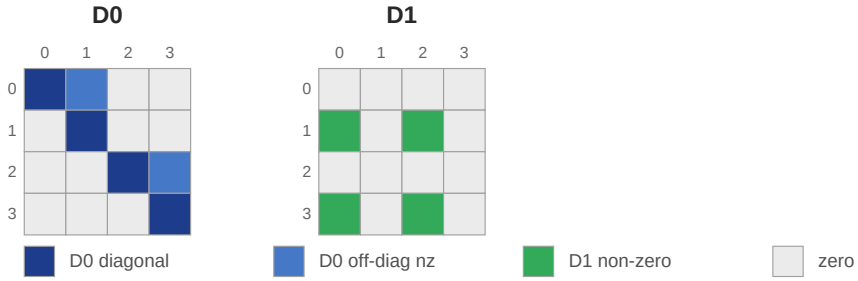
Description

k parallel Erlang chains, each with its own rate, and biased same-chain restart probabilities after each arrival. The Erlang phases keep SCV low (approx. 1/m_phases), while the same-chain restart bias creates positive autocorrelation. Fills the (rho1 > 0, SCV < 1) region that MMPPs and renewal processes cannot cover.

Matrix Structure

D0 is block-diagonal: within each Erlang chain, only the superdiagonal is non-zero (phase progression), all cross-chain entries are zero. D1 is sparse with a specific pattern: only the last phase of each chain can trigger an arrival; that row of D1 has non-zero entries only at the first phase of each chain (restart points), concentrated on the current chain (diagonal blocks dominant).

Sparsity Pattern (small example)



15. hyper_telegraph

Zone / Category: Zone C: negative rho1, high SCV

Autocorrelation (rho1): < -0.3 (negative) **SCV range: > 3 (high)**

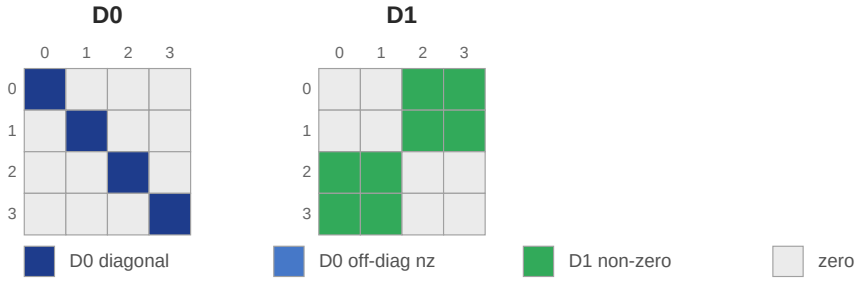
Description

Two blocks of hyperexponential states (fast and slow), with D1 strictly off-diagonal so every arrival switches the block. Fast-block rates are log-uniformly sampled around rate_ratio x base; slow-block rates around base. This guarantees rho1 < 0, and within-block rate spread drives SCV above 3.

Matrix Structure

D0 is block-diagonal: only the diagonal entries are non-zero within each block (hyperexponential = independent exponential states, no within-block transitions). All off-diagonal entries of D0 within each block and all cross-block entries are zero. D1 is block off-diagonal: upper-right (fast to slow) and lower-left (slow to fast) are dense; diagonal blocks of D1 are zero.

Sparsity Pattern (small example)



16. erlang_telegraph

Zone / Category: Zone D: negative rho1, low SCV

Autocorrelation (rho1): < 0 (negative) **SCV range: < 1 (low)**

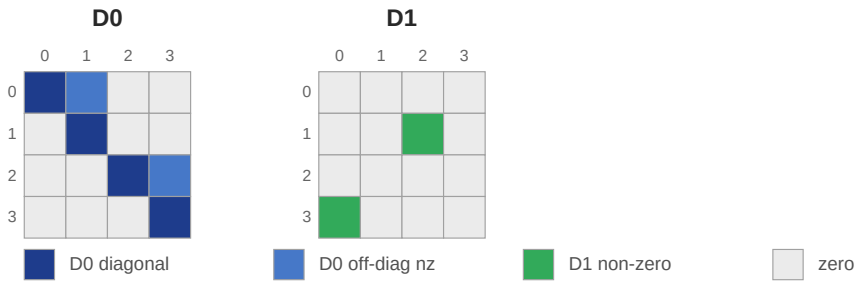
Description

Two Erlang chains with strictly off-diagonal D1: every arrival at the end of chain A restarts chain B and vice versa. The moderate rate ratio (1.2-8) keeps SCV well below 1 (Erlang phases reduce variability), while the alternation creates negative autocorrelation.

Matrix Structure

D0 is block-diagonal: within each Erlang chain, only the superdiagonal is non-zero (phase progression), all cross-chain entries are zero. D1 has exactly two non-zero entries: D1[last_A, first_B] = rate_a and D1[last_B, first_A] = rate_b. All other D1 entries are zero. This is the sparsest possible non-trivial D1 for a two-block MAP.

Sparsity Pattern (small example)



17. erlang_rate_spread

Zone / Category: Zone E: positive rho1, very high SCV

Autocorrelation (rho1): > 0.5 (positive) **SCV range: > 5 (very high)**

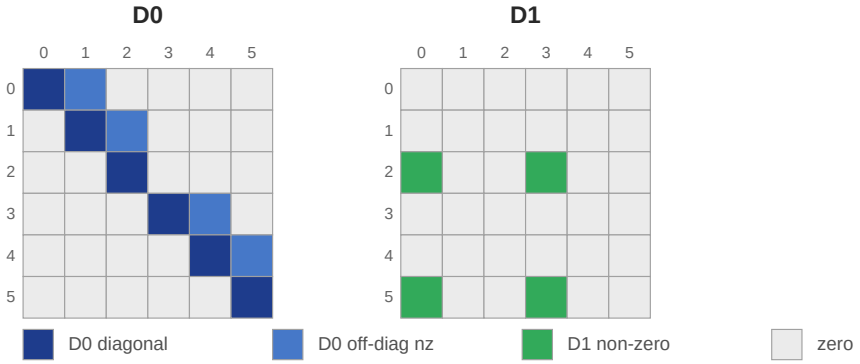
Description

k parallel Erlang chains with biased same-chain restart (like erlang_modulated), but with one rare slow anchor mode whose rate is rate_spread times smaller than the fast modes. The rare slow mode inflates SCV to O(1/weight), while the Erlang memory and biased D1 restart maintain rho1 > 0.5.

Matrix Structure

Same sparsity structure as erlang_modulated: D0 is block-diagonal with Erlang superdiagonals; D1 has non-zero entries only at the last phase of each chain, pointing to the first phase of each chain. The one structural difference from erlang_modulated is the magnitude: one chain has rates 1/rate_spread times smaller, making the slow mode rare but visible in the generated process statistics.

Sparsity Pattern (small example)



18. coxian_alternating

Zone / Category: Zone F: negative rho1, moderate SCV

Autocorrelation (rho1): < 0 (negative) **SCV range: Moderate to high**

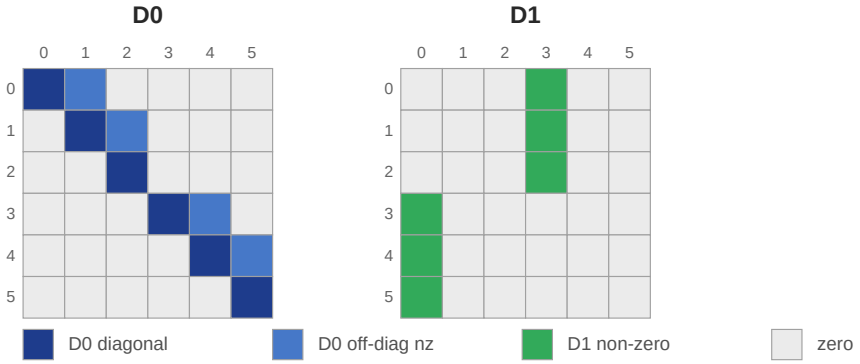
Description

Two Coxian PH blocks alternating strictly. Each block is a Coxian chain (forward progression with geometrically decaying phase-exit probabilities). Only the exits of block A send to block B and vice versa. The early-exit concentration creates a heavy-right-tail mixture (high SCV), while the alternation gives negative rho1.

Matrix Structure

D0 is block-diagonal: within each Coxian chain, the superdiagonal entries are non-zero (forward progression), and there are no cross-block entries. D1 is block off-diagonal: each phase i of block A has a non-zero entry D1[i, first_of_B] (exit to B start); each phase of B similarly exits to the start of A. Both off-diagonal blocks of D1 are column vectors (one non-zero column per block).

Sparsity Pattern (small example)



19. erlang_hyper_telegraph

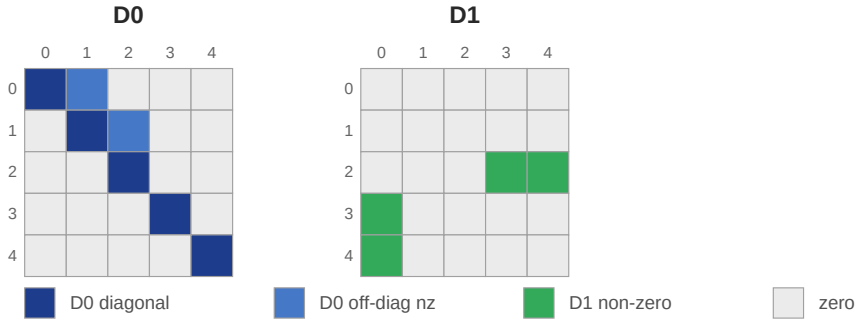
*Zone / Category: Boundary: rho1 * SCV approaches -1*

Autocorrelation (rho1): Near -1/SCV (theoretical minimum) **SCV range: 2 to 20+ (slides along boundary)**

Description

Erlang block A + Hyperexponential block B, strictly alternating D1. Analytically, as rate_ratio and k_erlang both grow, this structure achieves rho1 * SCV approaching -1, which is the theoretical minimum for any non-negative process. By varying hyper_spread (SCV of block B), the structure slides along the boundary curve rho1 = -1/SCV.

Matrix Structure

D0 block-diagonal: block A (Erlang) has superdiagonal-only non-zero entries; block B (hyperexponential) has diagonal-only non-zero entries (independent exponentials, no within-block transitions). D1 is block off-diagonal: only the last phase of A can trigger an arrival (row last_A of D1 is non-zero across all B columns with weights w_j); each phase of B triggers an arrival that restarts A at its first phase (D1[B_j, 0] non-zero for all j).

Sparsity Pattern (small example)